

Data Flow Diagram

Date	15 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A user or external system that interacts with the application, such as a Farmer or Weather API , by providing soil and environmental data or receiving crop recommendations.
Rectangle with numbered header	Process	An operation that processes the input data, such as Receive Soil Data (1.0) , Data Validation (2.0) , Crop Prediction using Machine Learning (3.0) , or Display Crop Recommendation (4.0) .
Rectangle (solid fill, no number)	Data Store	Stores agricultural data and machine learning resources used for prediction, such as Soil Data , Crop Database , Training Dataset , and Prediction Records .
Labelled arrow	Data Flow	Represents the movement of data between entities, processes, and data stores, such as Soil Data , Weather Data , Processed Data , and Crop Recommendations .

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

Data Flow Diagram (Level 0) – OptiCrop System



